

Authors	Origin	Basic equation	Viscosity variation	Solute size variation	Remarks
Sutherland (1905)	Parallel to Stokes-Einstein, but "no stick" at sphere's surface	$D = \frac{k_B T}{4\pi\mu R_0}$	μ^{-1}	R_0^{-1}	Always mentioned but rarely used
Glasstone et al. (1941)	Diffusion as a rate process	$D = \frac{k_B T}{2\mu R_0}$	μ^{-1}	R_0^{-1}	Closer to some experimental results
Scheibel (1954) ^{c,d}	Empirical	$D = \frac{AT}{\mu(\bar{V}_1)^{1/3}} \left[1 + \left(\frac{3\bar{V}_2}{\bar{V}_1} \right)^{2/3} \right]$	μ^{-1}	R_0^{-1} for large solutes and R_0^{-3} for small ones	Variation with solute size is the interesting feature
Wilke and Chang (1955) ^{e,f}	Empirical	$D = \frac{7.4 \cdot 10^{-8} (\phi \bar{M}_2)^{1/2} T}{\mu \bar{V}_1^{0.6}}$	μ^{-1}	Equivalent to $R_0^{-1.8}$	Factor ϕ for solute solvent interaction
King, Hsueh, and Mas (1965) ^g	Empirical	$D = 4.4 \cdot 10^{-8} \frac{T}{\mu} \left(\frac{\bar{V}_2}{\bar{V}_1} \right)^{1/6} \left(\frac{\Delta \bar{H}_{vap,2}}{\Delta \bar{H}_{vap,1}} \right)$	μ^{-1}	$R_0^{-0.5}$	Not suitable for viscous solvents or aqueous systems

Notes: ^aThe subscripts 1 and 2 indicate the solute and solvent, respectively.

^bThese relations are accurate within about ten percent for water and twenty percent for most organics, but they are often inaccurate for alcohols and other hydrogen-bonded solvents.

^cSpecific units implied are $D[=]\text{cm}^2/\text{sec}$; $T[=]\text{K}$; $\mu[=]10^{-2}\text{ g/cm-sec}$; $\bar{V}_1[=]\text{cm}^3/\text{g-mol}$.

^dThe $\bar{V}$ are the molar volumes at the boiling points. The constant A equals $8.2 \cdot 10^{-8}$, except as follows: $25.2 \cdot 10^{-8}$ for water if $\bar{V}_1 < \bar{V}_2$; $18.9 \cdot 10^{-8}$ for benzene when $\bar{V}_1 < 2\bar{V}_2$; $17.5 \cdot 10^{-8}$ for others if $\bar{V}_1 < 2.5\bar{V}_2$.

^eThe factor ϕ has the following values: 2.26 for water, 1.9 for methanol, 1.5 for ethanol, and 1.0 for non-hydrogen-bonded solvents.

Diffusion Coefficient in Porous Media

$$N_i = -D_{i,eff} \nabla c_i = -\frac{\epsilon}{\tau} D_i \nabla c_i$$

The tortuosity (τ) is, the ratio of the actual distance a molecule must travel between two points by following the fluid channel to the straight-line distance between those points.

Millington-Quirk: $\tau = \epsilon^{-1/3}$

Bruggeman: $\tau = \epsilon^{-1/2}$

In some types of porous media, the (effective) diffusion coefficient may also be **anisotropic**, so that the rate of diffusion depends on the direction of the concentration gradient. In this case, the diffusivity is a tensor.

Diffusion coefficient for gases

1. Chapman & Enskog

$$D_{AB} = \frac{1.8583 \times 10^{-7} T^{3/2}}{P \sigma_{AB}^2 \Omega_{D,AB}} \left(\frac{1}{M_A} + \frac{1}{M_B} \right)^{1/2}$$

P = absolute pressure in atm

σ_{AB} = average collision diameter

$\Omega_{D,AB}$ = collision integral based on Lennard-Jones potential;

is a ratio giving the deviation of a gas with interaction as compared to a gas rigid, elastic spheres.

2. Fuller et al.

$$D_{AB} = \frac{1.00 \times 10^{-7} T^{1.75} \left(\frac{1}{M_A} + \frac{1}{M_B} \right)^{1/2}}{P \left[\sum v_A^{1/3} + \left(\sum v_B^{1/3} \right)^2 \right]}$$

Convection Mass Transfer

- When a fluid flowing outside a solid surface in forced convection motion, rate of convective mass transfer is given by:

$$N_A = k_c (c_{L1} - c_{Li}) \rightarrow \text{similar to } h(T_s - T_\infty)$$

- k_c - mass transfer coefficient (m/s)
- c_{L1} - bulk fluid conc.
- c_{Li} - conc of fluid near the solid surface

- k_c depend on:

1. system geometry
2. Fluid properties
3. Flow velocity

How do we measure diffusion coefficients experimentally ?

Extra reading

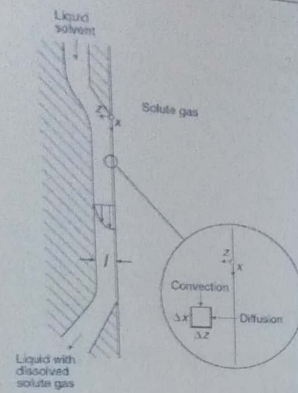
Mass transfer coefficients (MTC)

We would like to predict the mass transfer coefficient k as a function of the diffusion coefficient D and the fluid velocity v .

We look at MTC's for fluid-fluid systems that are VERY important in industrial applications.

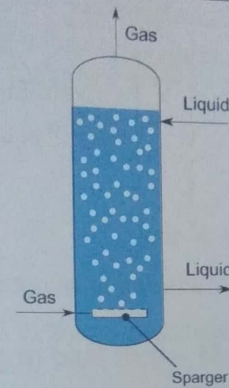
Fluid-Fluid Interfaces, e.g.

Falling film



Source: Cussler, Chapter 2.5.2

Gas bubbles in tank



Source: Wikipedia, "Blasensäule"

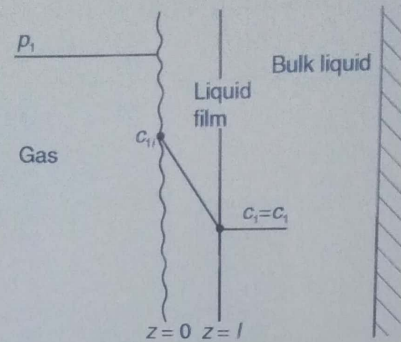
Liquid in packed tower



Source: Büchi Glas, Uster

Falling film	Gas bubbles in tank	Liquid in packed tower
Evaporation, Gas scrubbing	Aereation, Gas ab- and desorption, Bioreaktors	Extraction, Distillation
From Table 8.3-2: $\frac{kz}{D} = 0.69 \cdot \left(\frac{z \cdot v^0}{D} \right)^{0.5}$ $k \sim D^{0.5}, \sim v^{0.5}$	$\frac{k_d}{D} = 0.31 \cdot \left(\frac{d^3 \cdot g \cdot \Delta \rho / \rho}{v^2} \right)^{1/3} \left(\frac{v}{D} \right)^{1/3}$ $k \sim D^{2/3}$	$k \left(\frac{1}{vg} \right)^{1/3} = 0.0051 \left(\frac{v^0}{av} \right)^{0.67} \left(\frac{D}{v} \right)^{0.5} (ad)^{0.4}$ $k \sim D^{0.5}, \sim v^{0.67}$
Why is $k \sim D^{1/2}$ or $k \sim D^{2/3}$ or $k \sim v^{0.67}$??		

The Film Theory (Nernst, 1904)



Assumptions:

- All action (fluid flow and mass transfer) occurs in a thin film at the interface
- Bulk fluid (e.g. gas) – FILM – bulk fluid (e.g. liquid)
- Steady-state flux across film

$$n_1|_{z=0} = N_1 = k(c_{1i} - c_1) \quad (9.1-1)$$

This flux can be obtained also in terms of D
(for dilute concentrations)

$$N_1 = j_1|_{z=0} = \frac{D}{\ell}(c_{1i} - c_1) \quad (9.1-2)$$

Comparing equation (7.1) and (7.2) gives $k = \frac{D}{\ell}$ (9.1-3)

Or by rewriting gives $\frac{k\ell}{D} = 1 = Sh$ (9.1-4)

This simple theory gives $k \propto D^{1/2}$ BUT all fluid characteristics (e.g. fluid velocity due to stirring) are in the unknown film thickness ℓ .

This simple theory provides the FRAMEWORK of most MTC's as follows:

$$Sh = \frac{\left(\begin{array}{c} \text{mass transfer} \\ \text{coefficient} \end{array} \right) \left(\begin{array}{c} \text{characteristic} \\ \text{length} \end{array} \right)}{\left(\begin{array}{c} \text{diffusion} \\ \text{coefficient} \end{array} \right)} = F \left(\begin{array}{c} \text{other} \\ \text{system} \\ \text{variables} \end{array} \right)$$

Applications:

The film theory is used in some practical cases to determine the ℓ .

Example:

CO₂ is being scrubbed out of a gas by water flowing through a packed bed. Calculate the film thickness if $2.3 \cdot 10^{-6} \text{ mol}/(\text{cm}^2 \cdot \text{s})$ of CO₂ are adsorbed when

$$p_{\text{CO}_2} = 10 \text{ atm}, \quad H = 600 \text{ atm} \quad \text{and} \quad D_{\text{CO}_2/\text{H}_2\text{O}} = 1.9 \cdot 10^{-5} \text{ cm}^2/\text{s}.$$

Solution:

First find the interfacial concentration c_{1i} :

$$p_1 = H \cdot x_1 = H \frac{c_{1i}}{c}$$

$$10 \text{ atm} = 600 \text{ atm} \left(\frac{c_{1i}}{(1 \text{ mol})/(18 \text{ cm}^3)} \right) \Rightarrow$$

$$c_{1i} = 9.3 \cdot 10^{-4} \text{ mol}/\text{cm}^3$$

Calculate k from equation (9.1-1):

$$N_1 = k(c_{1i} - c_1)$$



$$2.3 \cdot 10^{-6} \text{ mol}/(\text{cm}^2 \text{ s}) = k (9.3 \cdot 10^{-4} \text{ mol}/\text{cm}^3 - 0)$$

$$k = 2.5 \cdot 10^{-3} \text{ cm/s}$$

Now

$$\ell = \frac{D}{k} = \frac{1.9 \cdot 10^{-5} \text{ cm}^2 / \text{s}}{2.5 \cdot 10^{-3} \text{ cm}^2 / \text{s}} = 0.76 \cdot 10^{-2} \text{ cm}$$

Typically $\ell \cong 10^{-2} \text{ cm}$ ← VERY IMPORTANT

The Penetration Theory (Higbie, 1935)

Assumptions:

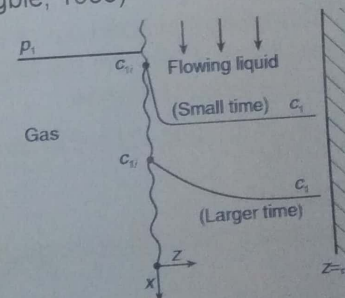
- Same as in "Film-theory" but the film is VERY thick
- Diffusion is important in z-direction
- Convection is important in x-direction

Equation (9.1-1): $N_1 = k(c_{1i} - c_1)$

The solution to this problem was given before in the context of the semi-infinite slab (Cussler p. 46):

$$N_1 = j_1|_{z=0} = \sqrt{D v_{\max}} / (\pi x) (c_{1i} - c_1) \quad (9.1-5)$$

where N_1 is the flux and $v_{\max}$ is the velocity of the liquid at the interface



Note that this flux at the interface is valid at a specific x . To find the average flux, $N_1(x)$ has to be averaged over the entire surface:

$$N_1 = \frac{1}{W \cdot L} \int_0^L \int_0^W n_1|_{z=0} dy \cdot dx$$

where L is the length of the film in x and W is its width in y . Since n_1 does not vary in y , inserting 9.1-5:

$$N_1 = \frac{1}{L} \int_0^L \sqrt{\frac{D \cdot v_{\max}}{\pi \cdot x}} \cdot \Delta c \, dx \quad \rightarrow \quad N_1 = \frac{2}{L} \sqrt{\frac{D \cdot v_{\max} \cdot x}{\pi}} \Big|_0^L \Delta c$$

$$N_1 = \frac{2}{L} \sqrt{\frac{D \cdot v_{\max} \cdot L}{\pi}} \cdot \Delta c$$

or
$$N_1 = 2 \sqrt{\frac{D \cdot v_{\max}}{\pi \cdot L}} \cdot (c_{ii} - c_1)$$

so
$$k = 2 \cdot \sqrt{\frac{D \cdot v_{\max}}{\pi \cdot L}} \quad (9.1-6)$$

The $L/v_{\max}$ is called contact time and is not known a priori in complex situations, as was ℓ in the film theory.

Compare: $k \propto D^{1/2}$ (penetration theory)
 $k \propto D$ (film theory)

These two theories bracket the experimental data (Table 8.3-2) very well, almost too well to be accepted.

Equation 9.1-6 can be rewritten, assuming that the average velocity is $v^0 = 2/3 v_{\max}$ (true for a laminar slit flow of a Newtonian fluid).

$$\begin{aligned} k \frac{L}{D} &= 2 \frac{L}{D} \left(\frac{3}{2} \right)^{1/2} \left(\frac{D v^0}{\pi L} \right) = \left(\frac{6}{\pi} \right)^{1/2} \left(\frac{L v^0}{D} \right)^{1/2} \\ &= \left(\frac{6}{\pi} \right)^{1/2} \left(\frac{L v^0}{v} \right)^{1/2} \left(\frac{v}{D} \right)^{1/2} \\ &= \left(\frac{6}{\pi} \right)^{1/2} \underbrace{\text{Re}^{1/2} \text{Sc}^{1/2}}_{\text{Pe}^{1/2}} \end{aligned}$$

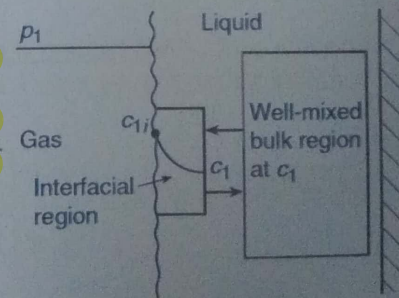
The Surface Renewal Theory (Dankwerts, 1951)

The success of the penetration theory with data despite its restrictive assumption motivated scientists and engineers to propose alternative and more realistic models leading, however, to the same variable dependencies at the end.

Assumption:

The thick film framework is replaced by TWO regions: interface and bulk.

In the interfacial region mass transfer takes place according to penetration theory. Then elements of this region ARE EXCHANGED with the bulk region. This is the so-called surface-renewal process.



The issue is how long the fluid elements stay in the interfacial region "exposed to penetration".

$$E(t)dt = \left(\begin{array}{c} \text{probability of a surface} \\ \text{element to be at the} \\ \text{surface for time } t \end{array} \right)$$

$E(t)$ is the residence time distribution, RTD, and $\int_0^{\infty} E(t)dt = 1$

The transfer of interfacial elements into the bulk is random and any surface element is equally likely to be withdrawn.

By definition the fraction of surface elements at time t is :

$$\theta = \exp[-t / \tau]$$

where τ is a characteristic constant equivalent to the average residence time of an element in the interfacial (surface) region.

Now the fraction θ is also the sum of probabilities $\theta = \int_t^{\infty} E(t)dt$

Thus the residence time distribution of surface elements is:

$$\frac{d}{dt} \left(\int_t^{\infty} E(t)dt \right) = \frac{d\theta}{dt} = -\frac{1}{\tau} \exp\left(-\frac{t}{\tau}\right) = \frac{d}{dt} \left(\underbrace{\int_0^{\infty} E(t)dt}_{=1} - \int_0^t E(t)dt \right) = -E(t)$$

So, $E(t) = \frac{1}{\tau} \exp\left(-\frac{t}{\tau}\right)$

The mass transfer coefficient at the interfacial region is obtained from the semi-infinite slab model (see eq. 9.1-5 and Cussler p. 46) as:

$$n_1|_{z=0} = N_1 = \sqrt{\frac{D}{\pi t}} \cdot (c_{1i} - c_1)$$

The semi-infinite slab model is used here even though the interfacial region is not infinite. But if the surface is rapidly renewed and the τ is small, then the interfacial region appears as infinite.

The average flux (over all surface elements) is:

$$\begin{aligned} N_{1,av} &= \int_0^{\infty} E(t) \cdot n_1|_{z=0} dt \\ &= \int_0^{\infty} \left(\frac{\exp[-t/\tau]}{\tau} \cdot \sqrt{\frac{D}{\pi t}} \cdot (c_{1i} - c_1) \right) dt \\ &= \sqrt{\frac{D}{\pi}} \frac{(c_{1i} - c_1)}{\tau} \cdot \int_0^{\infty} \frac{\exp[-t/\tau]}{\sqrt{t}} dt \\ &= \sqrt{\frac{D}{\pi}} \frac{(c_{1i} - c_1)}{\tau} \cdot \left(\frac{\sqrt{\pi} \cdot \operatorname{erf}(\sqrt{t/\tau})}{\sqrt{1/\tau}} \right) \Bigg|_0^{\infty} \end{aligned}$$

$$N_{1,av} = \sqrt{\frac{D}{\pi}} \frac{(c_{1i} - c_1)}{\tau} \cdot \sqrt{\pi\tau} \cdot \left(\underbrace{\text{erf}\left(\sqrt{\infty/\tau}\right)}_{=1} - \underbrace{\text{erf}\left(\sqrt{0/\tau}\right)}_{=0} \right)$$

$$N_{1,av} = \sqrt{\frac{D}{\tau}} (c_{1i} - c_1)$$

Thus, $k = \sqrt{D/\tau}$

As in the penetration theory, here $k \propto D^{1/2}$

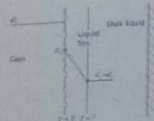
Again the residence time τ is as unknown as the ℓ in film theory or the $L/v_{\max}$ (contact time) in the penetration theory.

The major contribution of the surface renewal theory is that it gives a more REALISTIC physical situation. This gives a better starting point for development of effective correlations and better models.

Summary:

The Film Theory

$$\frac{k\ell}{D} = 1$$

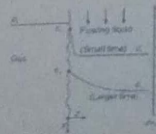


Simple;
good base for
extension

Film thickness ℓ
is unknown

The Penetration Theory

$$k = 2 \cdot \sqrt{\frac{D \cdot v_{\max}}{\pi \cdot L}}$$

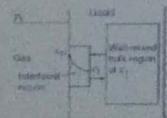


Simplest
including flow

Contact time
($L/v_{\max}$) usually
unknown

The Surface-Renewal Theory

$$k = \sqrt{D/\tau}$$



Similar math to
penetration theory,
but better physical
picture

Surface-renewal
rate (τ) is
unknown

The dependencies of k on D and v , like $k \sim D^{1/2}$, $k \sim D^{2/3}$ or $k \sim v^{0.67}$, observed in the experimentally-based MTCs are typically not well reflected by the simple mass transfer models.

These simple models for fluid-fluid interfaces pretend that fluid motion is incorporated in diffusion and everything is treated as a thin film or semi-infinite slab problem.

In principle, these two extreme cases should bracket all possible geometries. Yet, especially the effect of flow (velocity) is usually not well reflected.

One reason is that the simple theories assume a homogeneous system while real systems are heterogeneous with respect to concentration and flow (Schlünder, 1977).

Schlünder E.U., *Chem. Eng. Sci.* **32**, 845-851.